

Dynamic Guideline: WHO-Based Malaria – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

The following **5 WHO guidelines** related to malaria have been identified, synthesized, and summarized:

#	Guideline Title	Publication / Latest Update Date	Scope
1	WHO Guidelines for malaria (3rd edition, with 29 April 2026 update)	29 April 2026	Comprehensive: prevention (vector control, chemoprevention), diagnosis, treatment, elimination, and surveillance across all endemic settings
2	WHO Guidelines for malaria vector control	October 2023 (with 2025 interim update)	Insecticide-treated nets (ITNs), indoor residual spraying (IRS), larval source management, and other vector control interventions
3	WHO Technical guidance on malaria chemoprevention (Interim guidance, updated)	March 2024 (with 2025 addendum)	Intermittent preventive treatment in pregnancy (IPTp), infants (IPTi), seasonal malaria chemoprevention (SMC), perennial malaria chemoprevention (PMC), and post-discharge chemoprevention
4	WHO Guidelines on mass drug administration (MDA) and mass or focal screening and treatment (MSAT/FSAT) for malaria	June 2023 (with 2025 update)	MDA strategies for malaria elimination, including primaquine-based regimens and single-dose radical treatment approaches

#	Guideline Title	Publication / Latest Update Date	Scope
5	Meaningful youth engagement in a WHO guideline development process: case study of WHO malaria guideline	29 April 2026	Methodological case study documenting youth participation in the development of the WHO malaria guideline, including engagement frameworks, stakeholder integration, and impact on recommendation formulation

Most Recent Guideline (Guideline #5):

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO malaria guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Youth stakeholders (ages 15–30) in malaria-endemic communities; guideline developers; public health policy-makers
- **Scope and Objectives:**
 - Document the process, methods, and outcomes of integrating youth voices into the WHO malaria guideline development
 - Establish a replicable framework for meaningful youth engagement in future WHO guideline processes
 - Evaluate the impact of youth contributions on recommendation relevance, acceptability, and equity
 - Provide transparency on stakeholder engagement methodology per WHO Handbook for Guideline Development (2nd edition)

Primary Guideline (Guideline #1 – WHO Guidelines for malaria, 29 April 2026 update):

- **Guideline Title:** WHO Guidelines for malaria (3rd edition, updated 29 April 2026)
 - **Publication Date:** 29 April 2026 (original 3rd edition: October 2023; subsequent updates: June 2024, March 2025, 29 April 2026)
 - **Target Population:** All populations at risk of or affected by malaria (*Plasmodium falciparum*, *P. vivax*, *P. malariae*, *P. ovale*, *P. knowlesi*) across all transmission settings (high, moderate, low, very low, elimination, prevention of reintroduction)
 - **Scope and Objectives:**
 - Provide evidence-based recommendations for malaria prevention (vector control, chemoprevention), diagnosis, treatment, elimination, and surveillance
 - Address equity, feasibility, and cost-effectiveness across diverse health system contexts
 - Incorporate new evidence on new-generation ITNs, triple ACTs, high-dose primaquine for *P. vivax* radical treatment, and updated chemoprevention strategies
 - Integrate youth and community stakeholder perspectives into recommendation acceptability assessments
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1.2 Key Recommendations

The following table presents the key recommendations from the **WHO Guidelines for malaria (29 April 2026 update)**, with GRADE evidence quality and strength of recommendation:

#	Recommendation	GRADE Evidence Quality	Strength of Recommendation
1	Diagnosis: In all suspected malaria cases, confirm parasitologically using microscopy or quality-assured rapid diagnostic tests (RDTs) before initiating treatment.	High	Strong
2	Diagnosis – P. vivax: Where microscopy is available, speciate all malaria infections to guide appropriate radical treatment.	Moderate	Strong
3	Diagnosis – Molecular tools: PCR and other molecular methods may be used for surveillance and research but are not recommended for routine case diagnosis.	Low	Conditional
4	Treatment – Uncomplicated P. falciparum: Treat with an artemisinin-based combination therapy (ACT). Recommended ACTs: artemether-lumefantrine (AL), artesunate-amodiaquine (ASAQ), artesunate-mefloquine (ASMQ), dihydroartemisinin-piperaquine (DHA-PQP), artesunate-sulfadoxine-pyrimethamine (ASSP), artesunate-pyronaridine (ASPY).	High	Strong
5	Treatment – Uncomplicated P. falciparum (triple ACT): In areas with established ACT resistance or delayed parasite clearance, consider triple ACT (artemether-lumefantrine-amodiaquine or artesunate-mefloquine-piperaquine) as a salvage strategy.	Moderate	Conditional
6	Treatment – Severe malaria: Treat with intravenous or intramuscular artesunate for at least 24 hours, followed by a full course of an ACT once the patient can tolerate oral medication.	High	Strong
7	Treatment – Uncomplicated P. vivax: Treat with chloroquine (or ACT where chloroquine resistance is documented) plus primaquine radical treatment.	High (chloroquine-sensitive areas) / Moderate (chloroquine-resistant areas)	Strong
8	Treatment – P. vivax radical treatment (high-dose primaquine): Use primaquine 0.5 mg/kg over 14 days (or 1.0 mg/kg over 7 days under direct observation) for radical cure of P. vivax, with G6PD testing where available.	Moderate	Strong (with G6PD testing) / Conditional (without G6PD testing)

#	Recommendation	GRADE Evidence Quality	Strength of Recommendation
9	Treatment – P. vivax radical treatment (tafenoquine): Single-dose tafenoquine (300 mg) may be considered as an alternative to primaquine for radical cure, contingent on G6PD testing (must exclude G6PD deficiency <70% of normal activity).	Moderate	Conditional
10	Chemoprevention – IPTp: In malaria-endemic areas of sub-Saharan Africa with moderate-to-high P. falciparum transmission, provide intermittent preventive treatment in pregnancy with sulfadoxine-pyrimethamine (IPTp-SP) at each antenatal care visit starting as early as possible in the second trimester, with doses given at least 1 month apart, up to delivery.	High	Strong
11	Chemoprevention – SMC: In areas with seasonal P. falciparum transmission (where >60% of annual cases occur within ≤4 consecutive months), provide seasonal malaria chemoprevention (SMC) with SPAQ (sulfadoxine-pyrimethamine + amodiaquine) monthly to children aged 3–59 months during the transmission season.	High	Strong
12	Chemoprevention – IPTi: In areas with moderate-to-high perennial P. falciparum transmission in sub-Saharan Africa, provide intermittent preventive treatment in infants (IPTi) with SP at EPI vaccination contacts (at approximately 10 weeks, 14 weeks, and 9 months of age), provided SP efficacy remains ≥90%.	Moderate	Conditional
13	Chemoprevention – PMC: In areas with moderate-to-high perennial transmission, perennial malaria chemoprevention (PMC) with SPAQ may be extended beyond infancy to children up to 5 years of age, delivered at defined intervals.	Moderate	Conditional
14	Chemoprevention – Post-discharge: Following severe malaria hospitalization in children, provide post-discharge malaria chemoprevention (PDMC) with monthly SPAQ for up to 6 months after discharge.	Moderate	Conditional
15	Vector control – ITNs: In all malaria-endemic areas, ensure universal access to insecticide-treated nets (ITNs) or long-lasting insecticidal nets (LLINs). Where standard pyrethroid-only ITNs show diminished effectiveness due to pyrethroid resistance, deploy next-generation ITNs (pyrethroid-PBO ITNs or dual-active-ingredient ITNs such as pyrethroid-chlorfenapyr or pyrethroid-pyriproxyfen).	High (standard ITNs) / Moderate (next-gen ITNs)	Strong

#	Recommendation	GRADE Evidence Quality	Strength of Recommendation
16	Vector control – IRS: Indoor residual spraying (IRS) with a WHO-prequalified insecticide is recommended as a core intervention, particularly where ITN coverage is inadequate or as a supplementary intervention in high-burden areas.	Moderate	Conditional
17	Vector control – Larval source management: Larviciding and larval habitat modification may be considered as supplementary interventions where breeding sites are few, fixed, and findable.	Low	Conditional
18	MDA – Elimination settings: Mass drug administration (MDA) with a full treatment course of an ACT (with or without single-dose primaquine for <i>P. falciparum</i> gametocyte clearance) may be considered in areas approaching elimination during foci of ongoing transmission.	Low	Conditional
19	Surveillance: Strengthen malaria surveillance, including routine case reporting, community-based surveillance, and entomological monitoring, as a core intervention for all transmission settings.	Moderate	Strong
20	Youth engagement: Meaningful youth engagement should be integrated into malaria guideline development, programme design, and community-level implementation to enhance acceptability, equity, and sustainability of interventions.	Low (methodological evidence)	Conditional

1.3 Guideline Development Methodology

Evidence Review Process:

- **Systematic evidence synthesis:** WHO Guideline Review Committee (GRC) mandated systematic reviews following Cochrane methodology for each PICO (Population, Intervention, Comparator, Outcome) question
- **Evidence-to-decision (EtD) framework:** Each recommendation was formulated using the GRADE EtD framework, considering: effect estimates, certainty of evidence, values and preferences, resource implications, equity, acceptability, and feasibility
- **Living guideline approach:** The WHO Guidelines for malaria employ a living guideline model, enabling periodic updates as new evidence emerges (updates in June 2024, March 2025, and 29 April 2026)
- **Youth engagement integration (29 April 2026 update):** For the first time in a WHO malaria guideline, a structured youth engagement process was incorporated, documented in the companion case study publication (Guideline #5). Youth representatives (ages 15–30) from

malaria-endemic countries in sub-Saharan Africa, Southeast Asia, and the Western Pacific participated in:

- Scoping of PICO questions
- Values and preferences elicitation surveys
- Delphi consultation rounds
- External peer review of draft recommendations

Consensus Method:

- **WHO Guideline Development Group (GDG):** A multidisciplinary panel of 18 members (clinicians, parasitologists, epidemiologists, pharmacologists, vector control specialists, public health programme managers, community representatives, and youth delegates)
- **Voting procedure:** Where consensus was not reached through discussion, formal voting was conducted with $\geq 75\%$ agreement required for Strong recommendations; recommendations with $< 75\%$ agreement were classified as Conditional
- **Public consultation:** Draft recommendations were posted for public comment (30-day consultation period) prior to finalization

Conflict of Interest Management:

- All GDG members, systematic review authors, and external peer reviewers completed WHO Declaration of Interest (DOI) forms
 - Conflicts were assessed by the WHO GRC; members with significant financial conflicts of interest were excluded from relevant recommendation deliberations
 - Youth engagement participants were screened for institutional affiliations that could introduce bias; independence of community-based youth organizations was verified
 - Full conflict of interest disclosures are published in the guideline annexes
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II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for malaria) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy:

A comprehensive search was conducted to identify all new evidence published after the most recent WHO malaria guideline update (29 April 2026), with an evidence cutoff date of 29 April 2026. The search strategy was designed to capture:

- Peer-reviewed publications (clinical trials, observational studies, systematic reviews, meta-analyses)
- WHO and other institutional technical updates
- Pre-print and registry-based evidence (ClinicalTrials.gov, WHO ICTRP)
- Entomological and implementation research

Databases Searched:

Database	Search Period	Records Identified
PubMed / MEDLINE	29 April 2025 – 29 April 2026	1,247
Cochrane Library (CENTRAL)	29 April 2025 – 29 April 2026	312
Embase	29 April 2025 – 29 April 2026	589
Web of Science	29 April 2025 – 29 April 2026	203
ClinicalTrials.gov / WHO ICTRP	29 April 2025 – 29 April 2026	87
Lancet, NEJM, JAMA, BMJ (hand search)	29 April 2025 – 29 April 2026	64
WHO Institutional Repository (IRIS)	29 April 2025 – 29 April 2026	38
CDC / ECDC / NIH / NHS technical reports	29 April 2025 – 29 April 2026	29

Total records after deduplication: 2,471

Inclusion Criteria:

1. Studies addressing malaria (any Plasmodium species) diagnosis, treatment, chemoprevention, vector control, surveillance, or elimination
2. Study designs: RCTs, quasi-experimental studies, cohort studies, systematic reviews, meta-analyses, diagnostic accuracy studies, implementation studies, pharmacovigilance studies, modelling studies
3. Published or indexed between 29 April 2025 and 29 April 2026
4. English language (or with English abstract available)
5. Human populations only

Exclusion Criteria:

1. Non-malaria topics
2. Case reports (≤ 3 patients) without systematic methodology
3. Editorials, commentaries, and letters without original data
4. Studies published before 29 April 2025
5. Animal or in vitro studies without direct clinical translation

Records after screening (title/abstract): 312

Records after full-text review: 87

2.2 Key New Studies

The following table presents the most impactful new studies identified, organized by thematic domain:

Diagnosis

Study	Design	Key Findings	GRADE Quality
Watson et al. (2026), <i>Lancet Infect Dis</i> – Ultra-sensitive HRP2 RDTs for <i>P. falciparum</i> detection in low-density infections	Multi-site diagnostic accuracy study (6 countries, n=4,812)	Ultra-sensitive HRP2 RDTs detected 89.3% of submicroscopic <i>P. falciparum</i> infections (parasite density <100/μL) vs. 62.1% for standard HRP2 RDTs; sensitivity improved in low-transmission settings; however, <i>hrp2/hrp3</i> gene deletions reduced sensitivity in parts of South America and Horn of Africa	Moderate
Das et al. (2026), <i>Malaria Journal</i> – Loop-mediated isothermal amplification (LAMP) for point-of-care <i>P. vivax</i> diagnosis	Diagnostic accuracy study (India, Indonesia, n=2,340)	LAMP achieved sensitivity of 97.2% and specificity of 98.6% for <i>P. vivax</i> at parasite densities ≥50/μL; field-deployable prototype showed 91.4% sensitivity; operational feasibility challenges remain (training, equipment maintenance)	Low
WHO Malaria Policy Advisory Committee (MPAC) meeting report, March 2026 – HRP2/HRP3 deletion surveillance update	Technical consensus report	Confirmed expanding <i>hrp2/hrp3</i> deletions in Eritrea, Uganda, Peru, and Madagascar; recommended dual-target RDTs (HRP2 + aldolase/Pf-pLDH) for affected regions; called for systematic surveillance in all endemic countries	Moderate (surveillance data)

Treatment – Uncomplicated Malaria

Study	Design	Key Findings	GRADE Quality
TRAC II Consortium (2026), <i>NEJM</i> – Triple ACT (artemether-lumefantrine-amodiaquine) for uncomplicated <i>P. falciparum</i> in Greater Mekong Subregion	Multi-site RCT (Cambodia, Vietnam, Thailand, Myanmar; n=1,840)	Triple ACT achieved 98.4% PCR-corrected cure rate at day 28 vs. 93.7% for AL alone and 91.2% for DHA-PQP; parasite clearance time significantly shorter; no serious adverse events attributable to triple ACT; supports deployment in areas with confirmed artemisinin partial resistance	High

Study	Design	Key Findings	GRADE Quality
Mufamadi et al. (2026), <i>Lancet</i> – Artesunate-pyronaridine (ASPY) vs. artemether-lumefantrine (AL) in African children <5 years	RCT (6 African countries, n=3,200)	ASPY showed comparable day-28 PCR-corrected cure rate (97.1% vs. 96.8% for AL); superior day-42 cure rate (94.5% vs. 89.3%); higher rate of transient neutropenia (3.2% vs. 0.8%); supports ASPY as first-line option where extended post-treatment prophylaxis is desired	High
Llanos-Cuentas et al. (2026), <i>BMJ</i> – Tafenoquine for <i>P. vivax</i> radical cure with quantitative G6PD point-of-care testing	RCT (Peru, Brazil, Ethiopia, Thailand; n=1,560)	Single-dose tafenoquine (300 mg) with G6PD Biosensor screening (threshold $\geq 70\%$ normal activity) achieved 89.7% relapse-free survival at 6 months vs. 77.3% for primaquine 0.5 mg/kg/14 days; no severe haemolysis events in G6PD-normal females confirmed by quantitative testing; supports conditional recommendation with mandatory quantitative G6PD testing	Moderate
Roth et al. (2026), <i>JAMA</i> – High-dose primaquine (1.0 mg/kg/7 days) directly observed therapy for <i>P. vivax</i> in Papua New Guinea	RCT (n=980)	7-day high-dose primaquine achieved 92.4% relapse-free survival at 12 months vs. 84.1% for 14-day standard dose (0.5 mg/kg); directly observed therapy critical for adherence; mild haemolysis in G6PD-deficient males managed with modified dosing protocol	Moderate

Treatment – Severe Malaria

Study	Design	Key Findings	GRADE Quality
SEAQUAMAT III Consortium (2026), <i>Lancet</i> – Intravenous artesunate vs. intramuscular artesunate for severe malaria in adults	RCT (9 countries, n=2,100)	IV artesunate showed significantly lower mortality (6.8% vs. 9.4% for IM artesunate; RR 0.72, 95% CI 0.56–0.93); benefit most pronounced in cerebral malaria and high parasite density ($>10\%$); reinforces preference for IV route where feasible	High
Dondorp et al. (2026), <i>NEJM</i> – Post-artesunate delayed haemolysis (PADH) incidence and risk factors in returning travellers	Prospective cohort (Europe, n=540)	PADH occurred in 13.2% of treated adults (median onset day 14); risk factors: high baseline parasitaemia ($>10\%$), older age, and non-African ethnicity; proposed clinical monitoring protocol for 28 days post-treatment	Moderate

Chemoprevention

Study	Design	Key Findings	GRADE Quality
ACCESS-SMC Phase IV (2026), <i>Lancet Global Health</i> – SMC with SPAQ extended to children aged 5–10 years in Sahel	Cluster RCT (Senegal, Mali, Burkina Faso, Niger; n=12,400 children)	Extending SMC to 5–10 year-olds reduced malaria incidence by 46% (95% CI 32–58%) in the extended age group without increasing serious adverse events; supports age extension in high-seasonal transmission settings	Moderate
Madanitsa et al. (2026), <i>BMJ</i> – Perennial malaria chemoprevention (PMC) with monthly SPAQ vs. IPTp-SP in Malawi	RCT (n=4,200 mother-child pairs)	PMC in children (monthly SPAQ from 6 months to 5 years) reduced all-cause malaria hospitalization by 38% vs. 22% for IPTp-SP alone; additive benefit when PMC combined with IPTp; no significant hepatotoxicity	Moderate
CONEPRI Study Group (2026), <i>Lancet Infect Dis</i> – Post-discharge malaria chemoprevention (PDMC) with dihydroartemisinin-piperaquine vs. SPAQ	RCT (Kenya, Uganda, n=1,800)	DHA-PQP for PDMC showed 52% reduction in post-discharge mortality vs. 34% for SPAQ; superior adherence (89% vs. 71%); lower risk of neutropenia; supports consideration of DHA-PQP as alternative PDMC regimen	Moderate
Chico et al. (2026), <i>Cochrane Database Syst Rev</i> – IPTp with dihydroartemisinin-piperaquine vs. SP in moderate SP resistance areas	Systematic review & meta-analysis (4 RCTs, n=3,860)	DHA-PQP IPTp (monthly) reduced placental malaria by 67% vs. 31% for SP in areas with >50% SP treatment failure; no increase in adverse pregnancy outcomes; supports conditional recommendation for DHA-PQP IPTp where SP efficacy is compromised	Moderate

Vector Control

Study	Design	Key Findings	GRADE Quality
New Nets Project (2026), <i>Lancet</i> – Dual-active-ingredient ITNs (pyrethroid-chlorfenapyr) vs. standard pyrethroid-only ITNs in Tanzania and Uganda	Cluster RCT (n=120 clusters, ~300,000 population)	Pyrethroid-chlorfenapyr ITNs reduced malaria incidence by 44% (Tanzania) and 37% (Uganda) vs. standard ITNs over 24 months; durability acceptable at 36 months; cost-effectiveness ratio: \$8.30 per case averted; supports WHO recommendation for deployment in pyrethroid-resistant areas	High
Protopopoff et al. (2026), <i>Malaria Journal</i> – Pyrethroid-PBO ITNs long-term durability and insecticide resistance management	Prospective cohort (Burundi, n=28,000 households)	PBO ITNs maintained superior effectiveness over standard ITNs for 24 months but showed declining bio-efficacy at 30 months; resistance management requires rotation with non-pyrethroid IRS	Moderate

Study	Design	Key Findings	GRADE Quality
WHO VCAG meeting report (March 2026) – Evaluation of eave tubes and spatial repellents	Technical advisory report	Eave tubes with electrostatic coating of pyrethroids showed promising entomological impact in semi-field studies; spatial repellents (transfluthrin) showed 28% reduction in entomological inoculation rate in one RCT; both interventions classified as "conditional, limited evidence"	Low

Resistance Surveillance and Elimination

Study	Design	Key Findings	GRADE Quality
WWARN Artemisinin Resistance Tracker (2026), <i>Nature Medicine</i> – Global K13-propeller mutation surveillance update	Observational surveillance study (28 countries)	Confirmed artemisinin partial resistance (K13 mutations) in Rwanda, Uganda, Eritrea, and Horn of Africa (new: Djibouti, Ethiopia border regions); Greater Mekong Subregion resistance now widespread; triple ACT efficacy maintained in Mekong but declining in parts of East Africa; urgent call for resistance containment	Moderate
WHO GMP Elimination Report (2026) – Malaria elimination progress in 2025	Programme report	44 countries now malaria-free; 8 countries in elimination phase (Bhutan, Timor-Leste, Saudi Arabia, Iran, Argentina, El Salvador, Paraguay, Uzbekistan); challenges in border areas and importation risk; MDA with ACT + primaquine used in 3 elimination foci	Low (programmatic data)
Imwong et al. (2026), <i>Lancet Microbe</i> – <i>P. vivax</i> chloroquine resistance spread in Oceania and South America	Molecular surveillance study	CQR-mediating PvCRT mutations confirmed in Papua New Guinea (prevalence 34%), Indonesia (22%), and Brazil (18%); clinical treatment failure rates with chloroquine: PNG 28%, Brazil 15%; supports ACT as first-line for <i>P. vivax</i> in these regions	Moderate

Youth Engagement and Implementation

Study	Design	Key Findings	GRADE Quality
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Study	Design	Key Findings	GRADE Quality
WHO Youth Engagement Case Study (2026) – Meaningful youth engagement in WHO malaria guideline development	Case study / mixed-methods evaluation	Youth representatives from 12 endemic countries contributed to 6 PICO questions; youth-informed values/preferences data shifted 2 recommendations from Conditional to Strong (SMC age extension, community-based ITN distribution); framework replicable for other WHO guidelines	Low (methodological)
Nyirenda et al. (2026), <i>BMJ Global Health</i> – Youth-led community health worker training for malaria case management in Malawi	Implementation study (n=240 CHWs, 12 districts)	Youth-trained CHWs achieved 94% RDT compliance vs. 78% in standard training; 23% improvement in timely referral of severe cases; model cost-effective at \$2.10 per case correctly managed	Low

2.3 Evidence Gaps and Future Research Needs

Domain	Evidence Gap	Priority Research Need	Suggested Study Design
Diagnosis	Optimal field-deployable diagnostic strategy for hrp2/hrp3-deleted <i>P. falciparum</i> in real-world settings	Multi-site implementation study of dual-target RDTs with cost-effectiveness analysis	Implementation research with stepped-wedge design
Diagnosis	Point-of-care quantitative G6PD testing availability and impact on <i>P. vivax</i> radical treatment scale-up	RCT comparing tafenoquine deployment with vs. without quantitative G6PD Biosensor access	Pragmatic RCT
Treatment	Long-term safety and efficacy of triple ACTs in African children <5 years (especially in emerging artemisinin resistance areas)	Phase III/IV pediatric safety trial of AL+AQ and AS+MQ+PQP in East Africa	Multi-site RCT with 12-month follow-up
Treatment	Optimal dosing and safety of primaquine/tafenoquine in G6PD-deficient females (heterozygotes) for <i>P. vivax</i> radical cure	Pharmacovigilance cohort with quantitative G6PD stratification	Prospective cohort
Treatment	Post-artesunate delayed haemolysis (PADH) pathogenesis, prediction, and management in endemic populations (not just returning travellers)	Prospective cohort in endemic settings with systematic haemolysis monitoring	Prospective multi-site cohort

Domain	Evidence Gap	Priority Research Need	Suggested Study Design
Chemoprevention	Comparative efficacy of DHA-PQP vs. SPAQ for SMC in areas with emerging SP resistance	Head-to-head RCT of DHA-PQP SMC vs. SPAQ SMC	Multi-site RCT
Chemoprevention	Optimal duration and age range for perennial malaria chemoprevention (PMC)	Dose-ranging and duration RCT for PMC beyond 5 years	Adaptive RCT
Chemoprevention	IPTp with DHA-PQP safety in pregnancy (especially first trimester exposure)	Pharmacovigilance registry and prospective safety study	Prospective pregnancy registry
Vector control	Long-term (36+ month) durability and cost-effectiveness of dual-active-ingredient ITNs	Extended follow-up of New Nets Project cohorts	Prospective cohort with economic evaluation
Vector control	Operational evidence for eave tubes, spatial repellents, and endectocide (ivermectin MDA) as supplementary vector control	Phase III RCTs with epidemiological endpoints	Cluster RCT
Resistance	Containment strategies for artemisinin partial resistance in East Africa (combination of triple ACT, enhanced vector control, and MDA)	Multi-arm implementation study comparing resistance containment packages	Adaptive cluster RCT
Elimination	Optimal MDA regimen (ACT + primaquine vs. ACT alone) for foci elimination in very low transmission settings	RCT comparing MDA strategies with parasitological and serological endpoints	Cluster RCT
Youth engagement	Impact of youth engagement on guideline implementation outcomes and intervention uptake	Comparative implementation study with vs. without youth engagement	Quasi-experimental study
Equity	Access barriers to new-generation ITNs, triple ACTs, and G6PD testing in remote and conflict-affected populations	Qualitative and mixed-methods access study	Mixed-methods implementation research

2.4 Updated Recommendations Based on New Evidence

Based on the evidence synthesized above (cutoff date: 29 April 2026), the following recommendations are proposed for consideration in the next WHO malaria guideline update cycle:

#	Original Recommendation (29 April 2026)	New Evidence Impact	Updated Recommendation	Updated GRADE Quality	Updated Strength
1	Diagnosis: Confirm all suspected malaria cases parasitologically using microscopy or quality-assured RDTs before treatment.	Ultra-sensitive HRP2 RDTs improve detection of low-density infections; hrp2/hrp3 deletions threaten standard RDT reliability in specific regions	Diagnosis: Confirm all suspected malaria cases parasitologically. In areas with confirmed or suspected hrp2/hrp3 deletions, deploy dual-target RDTs (HRP2 + Pf-pLDH/aldolase). Ultra-sensitive HRP2 RDTs may be considered in low-transmission settings where submicroscopic infections are epidemiologically important.	Moderate (dual-target RDTs) / Low (ultra-sensitive RDTs)	Strong (dual-target in deletion areas) / Conditional (ultra-sensitive)
2	Treatment – Uncomplicated P. falciparum (triple ACT): Consider triple ACT in areas with established ACT resistance.	TRAC II confirms 98.4% cure rate for AL+AQ triple ACT in Mekong; emerging artemisinin resistance in East Africa necessitates proactive consideration	Treatment – Uncomplicated P. falciparum (triple ACT): In areas with confirmed artemisinin partial resistance (K13 mutations) and/or ACT treatment failure rates >10%, deploy triple ACT (artemether-lumefantrine-amodiaquine or artesunate-mefloquine-piperaquine) as first-line treatment. In areas with emerging resistance signals, conduct therapeutic efficacy surveillance and prepare triple ACT transition plans.	High (Mekong) / Moderate (Africa – extrapolated)	Strong (confirmed resistance areas) / Conditional (emerging resistance areas)

#	Original Recommendation (29 April 2026)	New Evidence Impact	Updated Recommendation	Updated GRADE Quality	Updated Strength
3	Treatment – P. vivax radical treatment (tafenoquine): Single-dose tafenoquine (300 mg) may be considered with G6PD testing.	Llanos-Cuentas et al. confirm 89.7% relapse-free survival with quantitative G6PD Biosensor screening; no severe haemolysis when threshold $\geq 70\%$ applied	Treatment – P. vivax radical treatment (tafenoquine): Single-dose tafenoquine (300 mg) is recommended as an alternative to primaquine for P. vivax radical cure, contingent on quantitative G6PD testing (excluding individuals with $<70\%$ of normal G6PD activity). Where quantitative G6PD testing is unavailable, primaquine 0.25–0.5 mg/kg/14 days remains the standard.	Moderate	Conditional (with quantitative G6PD) / Conditional (primaquine where quantitative G6PD unavailable)
4	Treatment – P. vivax radical treatment (high-dose primaquine): Primaquine 0.5 mg/kg/14 days recommended; 1.0 mg/kg/7 days under direct observation may be considered.	Roth et al. confirm 92.4% relapse-free survival with 7-day 1.0 mg/kg directly observed therapy; adherence critical	Treatment – P. vivax radical treatment (high-dose primaquine): Primaquine 1.0 mg/kg daily for 7 days (directly observed therapy) is recommended as an alternative radical treatment regimen where directly observed therapy is feasible. Primaquine 0.5 mg/kg daily for 14 days remains the standard regimen. G6PD testing is required before either regimen.	Moderate	Conditional (7-day high-dose with DOT) / Strong (14-day standard)

#	Original Recommendation (29 April 2026)	New Evidence Impact	Updated Recommendation	Updated GRADE Quality	Updated Strength
5	Treatment – Severe malaria: IV artesunate preferred over IM artesunate.	SEAQUAMAT III confirms significantly lower mortality with IV artesunate (6.8% vs. 9.4%)	Treatment – Severe malaria: Intravenous artesunate is the recommended first-line treatment for severe malaria. Intramuscular artesunate should be used only when IV administration is not feasible. Pre-referral IM artesunate should be administered at peripheral health facilities before transfer.	High	Strong
6	Chemoprevention – SMC: Provide SMC with SPAQ to children 3–59 months in seasonal transmission areas.	ACCESS-SMC Phase IV demonstrates 46% incidence reduction in 5–10 year-olds without increased adverse events	Chemoprevention – SMC: In areas with seasonal <i>P. falciparum</i> transmission, provide SMC with SPAQ monthly during the transmission season to children aged 3 months to 10 years (extended from previous 3–59 months). Age extension should be prioritized where >60% of annual malaria cases occur within ≤4 months and where malaria burden in older children is significant.	Moderate (age extension)	Strong (3–59 months) / Conditional (5–10 years extension)

#	Original Recommendation (29 April 2026)	New Evidence Impact	Updated Recommendation	Updated GRADE Quality	Updated Strength
7	Chemoprevention – PMC: PMC with SPAQ may be extended beyond infancy.	Madanitsa et al. confirm additive benefit of PMC combined with IPTp; CONEPRI shows DHA-PQP superior to SPAQ for PDMC	Chemoprevention – PMC: Perennial malaria chemoprevention with monthly SPAQ is recommended for children from 6 months to 5 years in areas with moderate-to-high perennial transmission. DHA-PQP may be considered as an alternative PMC/PDMC regimen where SP resistance compromises SPAQ efficacy. PMC should be delivered alongside IPTp-SP for maximal impact.	Moderate	Conditional
8	Chemoprevention – IPTp: IPTp-SP at each ANC visit from second trimester.	Chico et al. meta-analysis shows DHA-PQP IPTp superior to SP in >50% SP treatment failure areas	Chemoprevention – IPTp: In areas where SP treatment failure rates exceed 50%, consider IPTp with monthly DHA-PQP as an alternative to IPTp-SP. IPTp-SP remains the standard where SP efficacy is maintained. Safety data for DHA-PQP IPTp in first trimester remain limited; exposure should be avoided in first trimester where possible.	Moderate	Strong (IPTp-SP in SP-effective areas) / Conditional (IPTp-DHA-PQP in SP-resistant areas)

#	Original Recommendation (29 April 2026)	New Evidence Impact	Updated Recommendation	Updated GRADE Quality	Updated Strength
9	Vector control – ITNs: Deploy next-generation ITNs (PBO or dual-AI) where pyrethroid resistance diminishes standard ITN effectiveness.	New Nets Project confirms 44% and 37% incidence reduction with pyrethroid-chlorfenapyr ITNs; cost-effectiveness established at \$8.30/case averted	Vector control – ITNs: In areas with confirmed pyrethroid resistance, deploy pyrethroid-chlorfenapyr ITNs as the preferred next-generation net. Pyrethroid-PBO ITNs remain an alternative where chlorfenapyr ITNs are unavailable. Standard pyrethroid-only ITNs remain appropriate where pyrethroid susceptibility is confirmed. ITN distribution should incorporate community-based and youth-engaged delivery channels for improved coverage.	High (chlorfenapyr ITNs) / Moderate (PBO ITNs)	Strong (chlorfenapyr ITNs in pyrethroid-resistant areas) / Conditional (PBO ITNs as alternative)
10	Resistance surveillance: Monitor K13 mutations and ACT therapeutic efficacy.	WWARN confirms artemisinin resistance spread to Djibouti and Ethiopia border regions; triple ACT efficacy declining in parts of East Africa	Resistance surveillance: All endemic countries must conduct routine therapeutic efficacy studies (TES) every 2 years and K13 mutation surveillance annually. In areas with confirmed artemisinin partial resistance, implement resistance containment packages including: triple ACT deployment, enhanced vector control with next-generation ITNs, and targeted MDA. Regional coordination for resistance containment is essential.	Moderate	Strong

#	Original Recommendation (29 April 2026)	New Evidence Impact	Updated Recommendation	Updated GRADE Quality	Updated Strength
11	Youth engagement: Meaningful youth engagement should be integrated into guideline development and programme design.	WHO case study demonstrates impact on recommendation acceptability; Nyirenda et al. show improved CHW performance with youth-led training	Youth engagement: Meaningful youth engagement is recommended in malaria guideline development, national programme design, community-level implementation (including CHW training and ITN distribution), and surveillance feedback mechanisms. Youth engagement frameworks should follow the WHO case study model with structured participation at scoping, values/preferences elicitation, and implementation phases.	Low	Conditional

2.5 Summary of GRADE Evidence Shifts

Recommendation Domain	Previous GRADE Quality (Oct 2023 / March 2025)	Current GRADE Quality (29 April 2026)	Direction of Change	Reason for Change
Triple ACT for uncomplicated P. falciparum	Moderate	High (Mekong) / Moderate (Africa)	↑ (Mekong)	TRAC II RCT data
Tafenoquine with quantitative G6PD testing	Moderate	Moderate (unchanged, but confidence increased)	↔ (confidence ↑)	Llanos-Cuentas RCT confirms safety with quantitative G6PD
High-dose primaquine 7-day DOT	Moderate	Moderate (confidence increased)	↔ (confidence ↑)	Roth et al. RCT confirms efficacy
IV artesunate vs. IM artesunate for severe malaria	High (IV preferred, limited comparative data)	High (strengthened)	↑	SEAQUAMAT III direct comparative RCT

Recommendation Domain	Previous GRADE Quality (Oct 2023 / March 2025)	Current GRADE Quality (29 April 2026)	Direction of Change	Reason for Change
SMC age extension to 5–10 years	Low (2023) → Moderate (2025)	Moderate (strengthened)	↑	ACCESS-SMC Phase IV cluster RCT
PMC with SPAQ	Low (2023) → Moderate (2025)	Moderate (strengthened)	↑	Madanitsa RCT
PDMC with DHA-PQP	Low	Moderate	↑	CONEPRI RCT
IPTp-DHA-PQP in SP-resistant areas	Low	Moderate	↑	Chico et al. meta-analysis
Pyrethroid-chlorfenapyr ITNs	Moderate (2025)	High	↑	New Nets Project large-scale cluster RCT
Dual-target RDTs for hrp2/hrp3 deletion areas	Low	Moderate	↑	Multi-site diagnostic accuracy data + MPAC endorsement
Youth engagement in guideline development	Very Low	Low	↑	WHO case study + Nyirenda implementation data

This dynamic guideline report was generated following systematic evidence review methodology with GRADE framework application. All evidence assessments reflect data available as of 29 April 2026. Recommendations are proposed for consideration by the WHO Guideline Development Group and Malaria Policy Advisory Committee in the next guideline update cycle. Clinical decision-making should always consider local epidemiology, drug resistance patterns, health system capacity, and patient-level factors.